Black Box API Testing Report

QuickCart REST API

Roll No	2024101063
API Base	`http://localhost:8080/api/v1`
Tools	`pytest`, `requests`
Server	`docker run -p 8080:8080 quickcart`
Date	2026-03-24
Results	**118 passed** **13 failed** 4 skipped

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1 Test Case Design

Each test case lists the **input**, **expected output** (per API documentation), and **justification** for inclusion. Test cases cover valid requests, invalid inputs, missing fields, wrong data types, and boundary values.

1.1 Module 1: Header Validation

Headers are enforced by a single middleware layer. Testing on representative endpoints validates the global behaviour.

ID	Input	Expected	Justification
H-01	`GET /admin/users` — no headers	401	Every request must include `X-Roll-Number`
H-02	`GET /profile` — no headers	401	User endpoints also require roll number
H-03	`GET /admin/users`, `X-Roll-Number: abc`	400	Non-integer roll number rejected
H-04	`GET /profile` — only `X-Roll-Number`	400	User endpoints require `X-User-ID`
H-05	`GET /profile`, `X-User-ID: abc`	400	Non-integer user ID rejected
H-06	`GET /profile`, `X-User-ID: 999999`	400	Non-existent user must return 400
H-07	`GET /admin/users` — only `X-Roll-Number`	200	Admin endpoints work without `X-User-ID`

1.2 Module 2: Admin Endpoints

ID	Input	Expected	Justification
A-01	`GET /admin/users`	200, correct schema	Verify user list and key names
A-02	`GET /admin/users/2`	200, `user_id=2`	Single user lookup
A-03	`GET /admin/users/999999`	404	Non-existent user
A-04	`GET /admin/products`	200, includes inactive	Admin sees all products
A-05	`GET /admin/carts`	200, list	Admin cart inspection
A-06	`GET /admin/orders`	200, list	Admin order inspection

ID	Input	Expected	Justification
A-07	`GET /admin/coupons`	200, list	Includes expired coupons
A-08	`GET /admin/tickets`	200, list	All support tickets
A-09	`GET /admin/addresses`	200, list	All addresses

1.3 Module 3: Profile

ID	Input	Expected	Justification
P-01	`GET /profile`	200, has `user_id`, `name`, `email`, `phone`	Schema probe + fetch
P-02	`PUT /profile` — valid name + phone	200	Valid update accepted
P-03	`name: "K"` (1 char)	400	Name < 2 chars rejected
P-04	`name`: 51 chars	400	Name > 50 chars rejected
P-05	`name`: exactly 2 chars	200	Boundary minimum accepted
P-06	`name`: exactly 50 chars	200	Boundary maximum accepted
P-07	`phone: "123"` (3 digits)	400	Phone must be exactly 10 digits
P-08	`phone: "97045abc03"`	400	Non-digit phone rejected
P-09	`phone`: 11 digits	400	Phone > 10 digits rejected

1.4 Module 4: Addresses

ID	Input	Expected	Justification
AD-01	`GET /addresses`	200, list	Returns address list
AD-02	`POST` — valid HOME address	200/201, full schema	Create + schema check
AD-03	`label: "SCHOOL"`	400	Only HOME, OFFICE, OTHER valid
AD-04	`label: "OFFICE"`	200/201	Valid label accepted
AD-05	`label: "OTHER"`	200/201	Valid label accepted
AD-06	`street`: 3 chars	400	Street < 5 chars rejected

ID	Input	Expected	Justification
AD-07	`street`: exactly 5 chars	200/201	Boundary minimum accepted
AD-08	`street`: exactly 100 chars	200/201	Boundary maximum accepted
AD-09	`street`: 101 chars	400	Street > 100 chars rejected
AD-10	`pincode`: 4 digits	400	Must be exactly 6 digits
AD-11	`pincode`: exactly 6 digits	200/201	Boundary: 6 digits accepted
AD-12	`pincode`: 7 digits	400	Pincode ≠ 6 digits rejected
AD-13	`city`: 1 char	400	City < 2 chars rejected
AD-14	`city`: 51 chars	400	City > 50 chars rejected
AD-15	1 address as default	Exactly 1 default	Single default enforced
AD-16	2nd address as default	1st flips to false	Only one default allowed
AD-17	`PUT` — update street	200, new value in response	Response must show updated data
AD-18	`PUT` — send label/city/pincode	Those fields unchanged	Cannot modify via PUT
AD-19	`DELETE /addresses/{id}` — valid	200/204	Deletion works
AD-20	`DELETE /addresses/999999`	404	Non-existent returns 404

1.5 Module 5: Products

ID	Input	Expected	Justification
PR-01	`GET /products` vs admin	No inactive IDs in public list	Only active products shown
PR-02	Price in public vs admin	Prices match exactly	Price must not be modified
PR-03	`GET /products/{id}`	200, correct ID	Single product lookup
PR-04	`GET /products/999999`	404	Non-existent returns 404
PR-05	`?category=Fruits`	All have `category=Fruits`	Category filter correct

ID	Input	Expected	Justification
PR-06	`?search=Apple`	All names contain "Apple"	Name search correct
PR-07	`?sort=price_asc`	Prices ascending	Sort ASC works
PR-08	`?sort=price_desc`	Prices descending	Sort DESC works

1.6 Module 6: Cart

ID	Input	Expected	Justification
C-01	`GET /cart`	200, has `cart_id, items, total`	Schema probe
C-02	Add valid product, `qty=1`	200/201	Valid add succeeds
C-03	Add item, `qty=0`	400	Zero quantity rejected
C-04	Add item, `qty=-1`	400	Negative quantity rejected
C-05	`product_id: 999999`	404	Non-existent product
C-06	`qty` > `stock_quantity`	400	Over-stock rejected
C-07	Add same product twice (1+2)	`qty=3`	Quantities accumulate
C-08	Add qty=3; check subtotal	`subtotal = qty × price`	Subtotal calculation
C-09	Add 2 products; check total	`total = sum(subtotals)`	Total calculation
C-10	`POST /cart/update, qty=1`	200	Boundary minimum update
C-11	`POST /cart/update, qty=0`	400	Zero update rejected
C-12	Remove existing item	200	Remove works
C-13	Remove item not in cart	404	Non-existent returns 404
C-14	`DELETE /cart/clear`	200/204, empty cart	Cart fully cleared

1.7 Module 7: Coupons

ID	Input	Expected	Justification
CU-01	`GET /admin/coupons`	200, schema	Schema probe
CU-02	Apply expired coupon	400	Expired coupons rejected
CU-03	Valid coupon, cart < min	400	Min cart value enforced
CU-04	PERCENT coupon	200, discount = total×%	Percent math correct
CU-05	FIXED coupon	200, flat discount	Fixed math correct
CU-06	Discount exceeds cap	Discount = cap	Max cap enforced
CU-07	`code: "INVALIDCODE999"`	400/404	Invalid coupon rejected
CU-08	Remove with none applied	200 or 400	Graceful handling

1.8 Module 8: Checkout

ID	Input	Expected	Justification
CH-01	`GET /admin/orders`	200, order schema	Schema probe
CH-02	Checkout empty cart	400	Empty cart rejected
CH-03	`payment_method: "CRYPTO"`	400	Invalid method rejected
CH-04	COD, small order	200, `status: PENDING`	COD starts PENDING
CH-05	WALLET	200, `status: PENDING`	WALLET starts PENDING
CH-06	CARD	200, `status: PAID`	CARD starts PAID
CH-07	COD, total > 5001	400	COD blocked above 5000
CH-08	Invoice after COD	`total = subtotal × 1.05`	GST 5% for COD
CH-09	Invoice after WALLET	`total = subtotal × 1.05`	GST 5% for WALLET
CH-10	Invoice after CARD	`total = subtotal × 1.05`	GST 5% for CARD

1.9 Module 9: Wallet

ID	Input	Expected	Justification
W-01	`GET /wallet`	200, balance field	Schema probe
W-02	Add `amount=100`	200/201	Valid top-up
W-03	Add `amount=0`	400	Must be > 0 (boundary)
W-04	Add `amount=100001`	400	Exceeds max of 100000
W-05	Add `amount=100000`	200	Boundary maximum accepted
W-06	Pay `amount=100`	200	Valid payment
W-07	Pay `amount=0`	400	Zero payment rejected
W-08	Pay amount > balance	400	Insufficient funds
W-09	Pay 50; GET balance before + after	`after = before - 50`	Exact deduction verified

1.10 Module 10: Loyalty Points

ID	Input	Expected	Justification
L-01	`GET /loyalty`	200, points field	Schema probe
L-02	Redeem 0 points	400	Must redeem ≥ 1 point
L-03	Redeem > owned points	400	Cannot exceed balance
L-04	Redeem 1 point	200	Valid redemption

1.11 Module 11: Orders

ID	Input	Expected	Justification
O-01	`GET /orders`	200, list with `order_id`	Schema probe
O-02	`GET /orders/{id}`	200, matching `order_id`	Single order lookup
O-03	Cancel pending order	200	Valid cancel
O-04	Cancel DELIVERED order	400	Delivered cannot be cancelled
O-05	Cancel order 999999	404	Non-existent order
O-06	`GET /orders/{id}/invoice`	200, has `subtotal` + `total`	Invoice schema

ID	Input	Expected	Justification
O-07	`invoice.total` vs `order.total`	Values match exactly	Invoice must match order
O-08	Stock before/after cancel	`after = before + qty`	Cancellation restores stock

1.12 Module 12: Reviews

ID	Input	Expected	Justification
R-01	`GET /products/{id}/reviews`	200, schema	Schema probe
R-02	`rating=3`, comment valid	200/201	Valid review accepted
R-03	`rating=0`	400	Below minimum of 1
R-04	`rating=6`	400	Above maximum of 5
R-05	`rating=1`	200/201	Boundary minimum accepted
R-06	`rating=5`	200/201	Boundary maximum accepted
R-07	`comment: ""`	400	Empty comment rejected
R-08	`comment`: 1 char	200/201	Boundary minimum accepted
R-09	`comment`: 200 chars	200/201	Boundary maximum accepted
R-10	`comment`: 201 chars	400	Exceeds 200 chars
R-11	Ratings 3 and 4; GET reviews	`average_rating = 3.5`	Decimal average, not floored
R-12	Product with no reviews	`average_ratir = 0`	Zero when no reviews

1.13 Module 13: Support Tickets

ID	Input	Expected	Justification
T-01	`GET /support/tickets`	200, schema	Schema probe
T-02	POST valid ticket	200/201, `status: OPEN`	New ticket starts OPEN
T-03	`subject`: 4 chars	400	Below minimum of 5 chars

ID	Input	Expected	Justification
T-04	`subject`: 5 chars	200/201	Boundary minimum accepted
T-05	`subject`: 100 chars	200/201	Boundary maximum accepted
T-06	`subject`: 101 chars	400	Exceeds 100 chars
T-07	`message: ""`	400	Empty message rejected
T-08	`message`: 1 char	200/201	Boundary minimum accepted
T-09	`message`: 500 chars	200/201	Boundary maximum accepted
T-10	`message`: 501 chars	400	Exceeds 500 chars
T-11	Verbatim message with special chars	Stored exactly	Message must not be modified
T-12	OPEN → IN_PROGRESS	200	Valid forward transition
T-13	IN_PROGRESS → CLOSED	200	Valid forward transition
T-14	OPEN → CLOSED (skip)	400	Skipping a state rejected
T-15	IN_PROGRESS → OPEN	400	Backward transition rejected
T-16	CLOSED → IN_PROGRESS	400	No exit from CLOSED
T-17	CLOSED → OPEN	400	No exit from CLOSED

2 Bug Report

The following **13 bugs** were discovered by the automated test suite and confirmed with reproducible requests. Each bug includes the endpoint, full request payload, expected result (per API documentation), and actual result observed.

BUG-01 Non-existent User ID Returns 404 Instead of 400

Endpoint tested: `GET /api/v1/profile`

```
Method:  GET
URL:     http://localhost:8080/api/v1/profile
Headers: X-Roll-Number: 2024101063
         X-User-ID: 999999
Body:    (none)
```

Expected result: `400` — "If the header value is not valid or the user does not exist, the server returns a 400 error."

Actual result observed: `404 Not Found`

BUG-02 Profile — Phone with Letters Accepted

Endpoint tested: `PUT /api/v1/profile`

```
Method:  PUT
URL:     http://localhost:8080/api/v1/profile
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    {"name": "Kate Wilson", "phone": "97045abc03"}
```

Expected result: `400` — "The phone number must be exactly 10 digits."

Actual result observed: `200 OK` — Server accepted a phone number containing letters without any validation.

BUG-03 Addresses — PUT Response Returns Stale Data

Endpoint tested: `PUT /api/v1/addresses/{id}`

```
Method:  PUT
URL:     http://localhost:8080/api/v1/addresses/3
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    {"street": "New Updated Lane", "is_default": false}
```

Expected result: `200` with `"street": "New Updated Lane"` in the response body.

Actual result observed: `200 OK` but response contains `"street": "Old Street Road"` (pre-update value). The API doc states the response must reflect updated data.

BUG-04 Products — Public Price Differs from Database Price

Endpoint tested: `GET /api/v1/products`

```
Method:  GET
URL:     http://localhost:8080/api/v1/products
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    (none)
```

Expected result: Prices in public endpoint match `/admin/products` exactly for all products.

Actual result observed: Product ID 8: public endpoint returns `"price": 100`, admin returns `"price": 95`. The public endpoint serves an incorrect price.

BUG-05 Cart — Zero Quantity Accepted on Add

Endpoint tested: `POST /api/v1/cart/add`

```
Method:  POST
URL:     http://localhost:8080/api/v1/cart/add
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    {"product_id": 1, "quantity": 0}
```

Expected result: `400` — "The quantity must be at least 1."

Actual result observed: `200 OK` — Zero quantity silently accepted.

BUG-06 Cart — Negative Quantity Accepted on Add

Endpoint tested: `POST /api/v1/cart/add`

```
Method:  POST
URL:     http://localhost:8080/api/v1/cart/add
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    {"product_id": 1, "quantity": -1}
```

Expected result: `400` — "Sending a negative number must be rejected."

Actual result observed: `200 OK` — Negative quantities accepted, corrupting cart state (observed `"total": -16`).

BUG-07 Cart — Subtotal Does Not Equal Quantity × Price

Endpoint tested: `GET /api/v1/cart`

```
Method:  GET
URL:     http://localhost:8080/api/v1/cart
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    (none)
```

Expected result: For `qty=3`, `price=120` ⇒ `subtotal = 360`.

Actual result observed: `subtotal: 104` — Does not match `quantity × unit_price`.

BUG-08 Cart — Total is Negative

Endpoint tested: `GET /api/v1/cart`

```
Method:  GET
URL:     http://localhost:8080/api/v1/cart
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    (none)
```

Expected result: `total = sum of all item subtotals` (must be non-negative).

Actual result observed: `"total": -16` — Negative cart total is impossible for a valid cart. Result of BUG-05/06 compounded with incorrect summation logic.

BUG-09 Coupons — Valid PERCENT Coupon Incorrectly Rejected

Endpoint tested: `POST /api/v1/coupon/apply`

```
Method:  POST
URL:     http://localhost:8080/api/v1/coupon/apply
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    {"coupon_code": "SAVE10"}
```

Expected result: `200` — Valid, non-expired PERCENT coupon applied when cart total exceeds minimum.

Actual result observed: `400` — Valid coupon rejected with no clear error.

BUG-10 Wallet — Payment Deducts Imprecise Amount

Endpoint tested: `POST /api/v1/wallet/pay`

```
Method:  POST
URL:     http://localhost:8080/api/v1/wallet/pay
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    {"amount": 50}
```

Expected result: `balance_after = balance_before - 50` exactly.

Actual result observed: Balance before: `312546.39`, after: `312496.19` — Deduction of 50.20 instead of 50.00. Floating-point arithmetic not rounded correctly server-side.

BUG-11 Orders — Cancellation Does Not Restore Stock

Endpoint tested: `POST /api/v1/orders/{id}/cancel`

```
Method:  POST
URL:     http://localhost:8080/api/v1/orders/3026/cancel
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    (none)
```

Expected result: `stock_after = stock_before + cancelled_qty` — "When an order is cancelled, all items are added back to product stock."

Actual result observed: Stock before: `62`, after: `62` — Unchanged. Inventory not updated on cancellation.

BUG-12 Reviews — Rating of 0 Accepted (Below Minimum)

Endpoint tested: `POST /api/v1/products/1/reviews`

```
Method:  POST
URL:     http://localhost:8080/api/v1/products/1/reviews
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    {"rating": 0, "comment": "bad"}
```

Expected result: `400` — "A review rating must be between 1 and 5."

Actual result observed: `200 OK` — Rating of 0 accepted, violating the minimum constraint.

BUG-13 Reviews — Rating of 6 Accepted (Above Maximum)

Endpoint tested: `POST /api/v1/products/1/reviews`

```
Method:  POST
URL:     http://localhost:8080/api/v1/products/1/reviews
Headers: X-Roll-Number: 2024101063
         X-User-ID: 2
Body:    {"rating": 6, "comment": "too high"}
```

Expected result: `400` — "A review rating must be between 1 and 5."

Actual result observed: `200 OK` — Rating of 6 accepted, violating the maximum constraint.

Bug Summary Table

Bug ID	Module	Endpoint	Severity	Category
BUG-01	Headers	`GET /profile`	Medium	Wrong status code
BUG-02	Profile	`PUT /profile`	High	Missing validation
BUG-03	Addresses	`PUT /addresses/{id}`	Medium	Stale response
BUG-04	Products	`GET /products`	High	Incorrect data
BUG-05	Cart	`POST /cart/add`	High	Missing validation
BUG-06	Cart	`POST /cart/add`	High	Missing validation
BUG-07	Cart	`GET /cart`	High	Wrong calculation
BUG-08	Cart	`GET /cart`	High	Wrong calculation
BUG-09	Coupons	`POST /coupon/apply`	High	Valid input rejected
BUG-10	Wallet	`POST /wallet/pay`	Medium	Float precision error
BUG-11	Orders	`POST /orders/{id}/cancel`	High	Business rule violated
BUG-12	Reviews	`POST /products/{id}/reviews`	High	Missing validation
BUG-13	Reviews	`POST /products/{id}/reviews`	High	Missing validation